DATA 308: Lab 11b

Poisson regression

Dylan Wikel

```
library(performance)
library(broom)
library(tidyverse)
```

Neurons may fire selectively for particular types of stimuli. A visual-selective neuron is a brain cell that fires or responds strongly to specific features of a visual scene—such as a particular orientation, edge, motion direction, color, or complete object. To understand whether a neuron is a visual-selective neuron, 20 trials were run for each of the five image categories: animal, fruit, kids, military, space. In each trial, the number of spikes (the number of times that the neuron fired) within a 1-second window was recorded.

Let's run the code below to read in and examine our data.

```
neuron_data <- read_csv(
  "https://www.ics.uci.edu/~zhaoxia/Data/chosen_neuron_data.csv",
  show_col_types = FALSE
)

glimpse(neuron_data)
```

```
Rows: 100
Columns: 10
$ neuron_id        <dbl> 643, 643, 643, 643, 643, 643, 643, 643, 643, 643, 643~
$ trial_number     <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16~
$ n_spikes         <dbl> 0, 0, 0, 0, 0, 4, 0, 0, 0, 0, 0, 0, 0, 0, 0, 5, 0, 0,~
$ image_categ      <chr> "Animal", "Kids", "Fruit", "Space", "Space", "Fruit",~
$ image_response   <chr> "Yes animal", "Not animal", "Not animal", "Not animal~
$ neuron_location  <chr> "Left Amygdala", "Left Amygdala", "Left Amygdala", "L~
$ subject_id       <chr> "P29CS", "P29CS", "P29CS", "P29CS", "P29CS", "P29CS",~
$ session_id       <chr> "CS29_70", "CS29_70", "CS29_70", "CS29_70", "CS29_70"~
```

```
$ session_date     <dttm> 2013-10-01 08:00:00, 2013-10-01 08:00:00, 2013-10-
01~
$ session_location <chr> "Cedars-Sinai Medical Center", "Cedars-Sinai Medical ~
```

Question 1

Fit a model regressing number of neuron spikes on image category.

Response

```
neuron_model <-
  glm(n_spikes ~ image_categ,
      data = neuron_data,
      family = "poisson")

tidy(neuron_model)
```

```
# A tibble: 5 x 5
  term                 estimate std.error statistic   p.value
  <chr>                   <dbl>     <dbl>     <dbl>     <dbl>
1 (Intercept)         -3.00e+ 0     1.000 -3.00e+ 0 0.00274
2 image_categFruit     4.28e+ 0     1.01   4.25e+ 0 0.0000216
3 image_categKids      1.10e+ 0     1.15   9.51e- 1 0.341
4 image_categMilitary  1.61e+ 0     1.10   1.47e+ 0 0.142
5 image_categSpace     1.48e-15     1.41   1.05e-15 1.000
```

Question 2

Write out the fitted regression model in context.

Response

$log(\widehat{\text{Neuron Spike}}_i) = -2.995732 + 4.276666(\text{Image Category}_i = \text{Fruit}) + 1.098612(\text{Image Category}_i = \text{Kids}) + 1.609438(\text{Image Category}_i = \text{Military}) + 1.479608 \times 10^{-15}(\text{Image Category}_i = \text{Space})$

Question 3

Identify the estimate and 95% confidence interval for β_j for the kids category. Provide an interpretation in context.

Response

$\beta_{\text{Kids}} = 1.09861\$$ is the estimated log rate ratio of neuron spikes when looking at an image in the kids category compared to looking at an image in the animal category, between -0.9568748 and 4.105099, with 95% confidence.

Question 4

Identify the estimate and 95% confidence interval for e^{β_j} for the kids category. Provide an interpretation in context.

Response

```
tidy(neuron_model, conf.int = TRUE, exponentiate = TRUE)
```

```
# A tibble: 5 x 7
  term                estimate std.error statistic   p.value conf.low conf.high
  <chr>                  <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl>
1 (Intercept)           0.0500     1.000 -3.00e+ 0 0.00274    0.00285     0.220
2 image_categFruit     72.0        1.01   4.25e+ 0 0.0000216 16.0       1271.
3 image_categKids       3.00       1.15   9.51e- 1 0.341      0.384       60.6
4 image_categMilitary   5.00       1.10   1.47e+ 0 0.142      0.806       95.8
5 image_categSpace      1.00       1.41   1.05e-15 1.000      0.0395      25.3
```

$e^{\beta_{kids}} = 3$ is the estimated rate ratio of neuron spikes when looking at an image in the kids category compared to looking at an image in the animal category, between 0.384091366 and 60.6487384, with 95% confidence

Question 5

Let's conduct a hypothesis test to determine if there is evidence that the expected number of spikes in a one-second window associated with image category?

Question 5a

What are the null and alternative hypotheses?

Response

H_0 : all $\beta's = 0$ H_a : at least one $\beta \neq 0$

Question 5b

Conduct the appropriate test and report you p-value with your statistical decision. Hint: There are no predictors for the reduced model, so you can just specify `n_spikes ~ 1` as the formula.

Response

```
partial_neuron_model <-
  glm(n_spikes ~ 1,
      data = neuron_data,
      family = "poisson")

anova(neuron_model, partial_neuron_model)
```

```
Analysis of Deviance Table

Model 1: n_spikes ~ image_categ
Model 2: n_spikes ~ 1
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1        95     81.213
2        99    260.985 -4  -179.77 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 5c

Report your conclusion in context for your hypothesis test.

Response

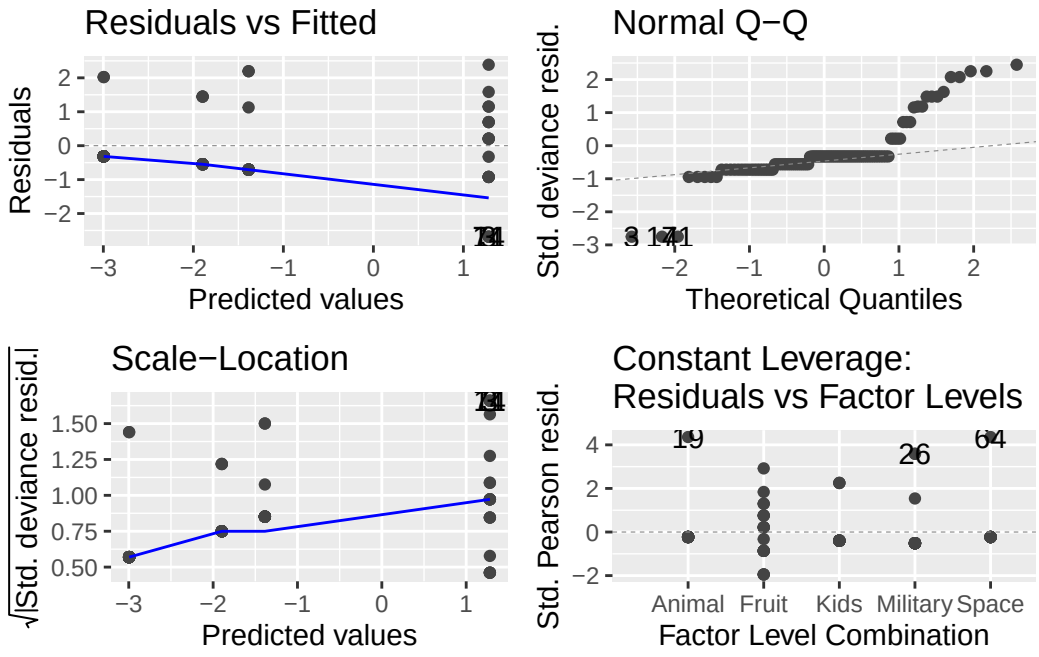
We found significant evidence $p < 2.2 \times 10^{-16}$. Therefore, we reject the null hypothesis and conclude that there is significant evidence that at least one image category is associated with a different spike rate in this particular neuron.

Question 6: Independence of observations condition

A model condition for every model we have learned is independence of observations. We have repeated trials from one neuron. Based on the information provided, can we establish that spike counts from different trials are independent? Identify one reason that counts could be related across trials and one additional detail about the experiment that would help you evaluate this condition.

Response

```
library(ggfortify)
autoplot(neuron_model)
```



Based on the information provided, we cannot establish that spike counts from different trials are independent. Since the same neuron was measured repeatedly, its response during one

trial could be influenced by previous stimulation. For example, the neuron might become less responsive after repeated exposure to certain images, which could affect subsequent spike counts. One additional detail that would help evaluate independence is the amount of time between trials. Knowing this would help us determine whether the neuron had enough time to recover between stimuli, although it would not guarantee independence.

Question 7: Scope of the conclusion

Do these 100 trials provide information about 100 different neurons? Explain how this may or may not limit generalization to many neurons.

Response

No, the trials provide information about one single neuron. The generalization may only be made about this particular neuron in this one participant.

Question 8 (Challenge your understanding) : Removing the intercept

The only variable we have in our original model is categorical, so we have $k-1$ indicators as predictors in our model. If we wanted to be able to interpret the relationship between an image category and the response variable without comparing to a reference group, we can remove the intercept by specifying `n_spikes ~ -1 + imgage_categ` as the formula.

Question 8a

Fit this model write out the fitted model.

Response

```
no_ref_model <-
  glm(n_spikes ~ -1 + image_categ,
      data = neuron_data,
      family = "poisson")

tidy(no_ref_model)
```

```
# A tibble: 5 x 5
  term                estimate std.error statistic  p.value
  <chr>                  <dbl>     <dbl>     <dbl>    <dbl>
1 image_categAnimal      -3.00     1.000     -3.00 2.74e- 3
2 image_categFruit        1.28     0.118     10.9  1.62e-27
3 image_categKids        -1.90     0.577     -3.29 1.02e- 3
4 image_categMilitary    -1.39     0.447     -3.10 1.94e- 3
5 image_categSpace       -3.00     1.000     -3.00 2.74e- 3
```

Question 8b

Identify and interpret your estimate for e^{β_j} for the kids predictor.

Response

```
tidy(no_ref_model, exponentiate = TRUE)
```

```
# A tibble: 5 x 5
  term                estimate std.error statistic  p.value
  <chr>                  <dbl>     <dbl>     <dbl>    <dbl>
1 image_categAnimal     0.0500     1.000     -3.00 2.74e- 3
2 image_categFruit      3.60       0.118     10.9  1.62e-27
3 image_categKids       0.150      0.577     -3.29 1.02e- 3
4 image_categMilitary   0.250      0.447     -3.10 1.94e- 3
5 image_categSpace      0.0500     1.000     -3.00 2.74e- 3
```

$e^{\beta_{kids}} = 0.15$ is the estimated rate ratio of neuron spikes when looking at an image in the kids category recorded within a one second time window.

Question 8c

How does your original model compare to this model? Are there any similarities?

Response

Fruit is consistently a larger estimate. By not comparing it to a reference group, it gives us a more interpretable value for the average rate of neuron spikes per second that corresponds to each image type. Another thing I noticed is that all of the p-values have become significant after modifying the model to not compare to the reference group. However, I think that they are now not in proper context for interpretation.